
Table of contents

1 accounts	Page number: 2
2 classes	Page number: 3
3 class_schedules	Page number: 4
4 courseclasses	Page number: 5
5 courses	Page number: 6
6 faculties	Page number: 7
7 grades	Page number: 8
8 rooms	Page number: 9
9 semesters	Page number: 10
10 students	Page number: 11
11 teachers	Page number: 12
12 Relational schema	Page number: 13

1 accounts

Creation: Jun 25, 2026 at 01:57 PM
Last update: Jun 25, 2026 at 03:46 PM

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
username	varchar(20)		No					
password	varchar(255))		Yes	NULL				
role	varchar(20)		No					
isActive	tinyint(1)		Yes	1				
createdAt	datetime		Yes	current_time stamp()				

2 classes

Creation: Jun 25, 2026 at 01:57 PM
Last update: Jun 25, 2026 at 03:35 PM

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
maLop	varchar(20)		No					
tenLop	varchar(100))		No					
email	varchar(100))		Yes	NULL				
maKhoa	varchar(20)		No			-> faculties.maKhoa ON UPDATE RESTRICT ON DELETE RESTRICT		
maGV	varchar(20)		No			-> teachers.maGV ON UPDATE RESTRICT ON DELETE RESTRICT		

3 class_schedules

Creation: Jun 25, 2026 at 01:57 PM

Last update: Jun 25, 2026 at 03:42 PM

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
schedule_id	int(11)		No		auto_increment			
day_of_week	tinyint(4)		No				2=Thứ 2, ..., 8=Chủ nhật	
start_period	tinyint(4)		No				Tiết bắt đầu (1-15)	
num_periods	tinyint(4)		No				Số tiết kéo dài	
start_date	date		Yes	NULL				
end_date	date		Yes	NULL				
maLHP	int(11)		No			-> courseclasses.maLHP ON UPDATE RESTRICT ON DELETE CASCADE		
room_id	int(11)		No			-> rooms.room_id ON UPDATE RESTRICT ON DELETE RESTRICT		

4 courseclasses

Creation: Jun 25, 2026 at 01:57 PM

Last update: Jun 25, 2026 at 03:41 PM

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
maLHP	int(11)		No		auto_increment			
maMH	varchar(20)		No			-> courses.maMH ON UPDATE RESTRICT ON DELETE RESTRICT		
maHK	varchar(20)		No			-> semesters.maHK ON UPDATE RESTRICT ON DELETE RESTRICT		
maGV	varchar(20)		No			-> teachers.maGV ON UPDATE RESTRICT ON DELETE RESTRICT		
isLocked	tinyint(1)		Yes	0				

5 courses

Creation: Jun 25, 2026 at 01:57 PM

Last update: Jun 25, 2026 at 03:35 PM

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
maMH	varchar(20)		No					
tenMH	varchar(100)		No					
soTinChi	int(11)		Yes	3				
maKhoa	varchar(20)		No			-> faculties.maKhoa ON UPDATE RESTRICT ON DELETE RESTRICT		

6 faculties

Creation: Jun 25, 2026 at 01:57 PM

Last update: Jun 25, 2026 at 01:57 PM

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
maKhoa	varchar(20)		No					
tenKhoa	varchar(100)		No					
diaChi	varchar(200)		Yes	NULL				
email	varchar(100)		Yes	NULL				
sdt	varchar(20)		Yes	NULL				

7 grades

Creation: Jun 25, 2026 at 01:57 PM

Last update: Jun 25, 2026 at 03:46 PM

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
maSV	varchar(20)		No			-> students.maSV ON UPDATE RESTRICT ON DELETE RESTRICT		
maLHP	int(11)		No			-> courseclasses.maLHP ON UPDATE RESTRICT ON DELETE RESTRICT		
diem1	decimal(4,2)		Yes	NULL				
diem2	decimal(4,2)		Yes	NULL				
diemThi	decimal(4,2)		Yes	NULL				
diemTong	decimal(4,2)		Yes	NULL	VIRTUAL GENERATED			
diemTongChu	varchar(2)		Yes	NULL	VIRTUAL GENERATED			

8 rooms

Creation: Jun 25, 2026 at 01:57 PM

Last update: Jun 25, 2026 at 01:57 PM

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
room_id	int(11)		No		auto_increment			
room_name	varchar(50)		No					
capacity	int(11)		No	50				
room_type	varchar(50)		Yes	Lý thuyết				

9 semesters

Creation: Jun 25, 2026 at 01:57 PM
Last update: Jun 25, 2026 at 01:57 PM

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
maHK	varchar(20)		No					
tenHK	varchar(50)		No					

10 students

Creation: Jun 25, 2026 at 01:57 PM
Last update: Jun 25, 2026 at 03:45 PM

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
maSV	varchar(20)		No					
hoTen	varchar(100)		No					
gioiTinh	varchar(10)		Yes	NULL				
namSinh	int(11)		Yes	NULL				
diaChi	varchar(200)		Yes	NULL				
email	varchar(100)		Yes	NULL				
sdt	varchar(20)		Yes	NULL				
maLop	varchar(20)		No			-> classes.maLop ON UPDATE RESTRICT ON DELETE RESTRICT		

11 teachers

Creation: Jun 25, 2026 at 01:57 PM
Last update: Jun 25, 2026 at 03:35 PM

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
maGV	varchar(20)		No					
hoTenGV	varchar(100))		No					
email	varchar(100))		Yes	NULL				
sdt	varchar(20)		Yes	NULL				

